

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS – GRADE 10 SUB-STRAND 2.3: ROTATION	
Student Assessment Document	
Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation document. Use it as a model to help you write a complete, well-reasoned answer to the driving question. Read each section carefully, study the exemplar responses, then write your own answer using the prompts as a guide. Your goal is to show that you truly understand rotation — not just the steps, but the "why" behind every rule. Refer back to the University of Nairobi clock tower phenomenon as you write: every idea in this document connects to what you observed about that minute hand.

Section 1 — What Is Rotation? Defining the Four Pillars

In your own words, define rotation as a geometric transformation. Your answer must name and explain all four pillars: the centre of rotation, the angle of rotation, the direction of rotation, and the fact that shape and size are preserved. Use the University of Nairobi clock tower as your anchor example.

Rotation is a transformation that turns every point of a shape through a fixed angle around one special fixed point called the centre of rotation. Think about the minute hand on the clock at the University of Nairobi Administration Block. As it moves from 12:00 to 12:15, four things are always true — and these four truths are the pillars of every rotation:

1. CENTRE OF ROTATION: There is one fixed point that does not move at all. On the Nairobi clock, this is the pivot at the very centre of the clock face. On a coordinate grid, it is most often the origin (0, 0), but it can be any point. Every other point on the shape turns around this centre.
2. ANGLE OF ROTATION: Every point on the shape turns through exactly the same angle. The minute hand turns 90° every 15 minutes (a full 360° in 60 minutes). We always measure this angle from the original position (the object) to the new position (the image). Common angles you will meet are 90° , 180° , and 270° .
3. DIRECTION OF ROTATION: Rotation can go clockwise (the same direction as the clock hands — from 12 toward 3 toward 6) or anticlockwise (the opposite direction — from 12 toward 9 toward 6). You must always state the direction, because a 90° clockwise rotation and a 90° anticlockwise rotation land the hand in completely different places.
4. PRESERVATION OF SHAPE AND SIZE (ISOMETRY): After rotation, the image is exactly the same shape and size as the original object. The minute hand does not stretch, shrink, or flip — it simply points in a new direction. In mathematics, we say rotation is an isometric transformation: all lengths and all angles within the shape are preserved.

Section 2 — The Mathematical Rules: Rotation Formulae on a Coordinate Grid

Explain the rotation rules for 90° clockwise, 90° anticlockwise, and 180° about the origin. Show clearly how each rule changes the coordinates of a point. Then apply one rule to a real example using coordinates you choose yourself.

When a shape sits on a coordinate grid and you rotate it about the origin (0, 0), you can calculate the exact new position of every point using coordinate rules. These rules come directly from the four pillars: the centre is (0, 0), the shape is preserved, and only the coordinates change.

Imagine the tip of the Nairobi clock's minute hand is at the coordinate point P(3, 4) when the time is 12:00.

RULE 1 — 90° ANTICLOCKWISE (same as 90° clockwise on a clock face, but in geometry anticlockwise is the positive/default direction):

$$(x, y) \rightarrow (-y, x)$$

So P(3, 4) $\rightarrow$ P'(-4, 3). The tip of the hand has moved to the left and slightly upward — exactly what you would see if the hand moved from the 12 toward the 9.

RULE 2 — 90° CLOCKWISE:

$$(x, y) \rightarrow (y, -x)$$

So P(3, 4) $\rightarrow$ P'(4, -3). The tip moves to the right and downward — the hand has moved from the 12 toward the 3 and then toward the 6.

RULE 3 — 180° (either direction — the result is the same):

$$(x, y) \rightarrow (-x, -y)$$

So P(3, 4) $\rightarrow$ P'(-3, -4). The hand now points in the completely opposite direction — exactly as it would at 12:30 compared to 12:00.

WHY DO THESE RULES WORK? Each rule is derived from trigonometry (specifically from the general rotation formula using sine and cosine), but for the special angles 90° and 180°, the sine and cosine values simplify to 0, 1, and -1, giving us these clean, easy rules.

IMPORTANT: These rules apply when the centre of rotation is the origin. If the centre is a different point, say C(a, b), you must first translate the shape so that C moves to the origin, apply the rule, then translate back.

Section 3 — Rotating a Full Shape: A Step-by-Step Method

Describe the full method for rotating a triangle (or any polygon) on a coordinate grid. Walk through every step clearly. Include how to plot the object, apply the rotation rule to each vertex, plot the image, and

To rotate a full shape, you apply the rotation rule to every vertex (corner point) of the shape, one at a time. Here is the complete method, using triangle KNB — named after Kisumu, Nairobi, and Baringo — with vertices K(1, 2), N(4, 2), and B(4, 5). We will rotate it 90° anticlockwise about the origin.

STEP 1 — DRAW AND LABEL THE OBJECT.

Plot points K(1, 2), N(4, 2), and B(4, 5) on your grid. Join them to form triangle KNB. This is your object.

label it correctly. Use a triangle with Kenyan-context coordinates as your example.	STEP 2 — APPLY THE ROTATION RULE TO EACH VERTEX. Using the rule for 90° anticlockwise: $(x, y) \rightarrow (-y, x)$ – $K(1, 2) \rightarrow K'(-2, 1)$ – $N(4, 2) \rightarrow N'(-2, 4)$ – $B(4, 5) \rightarrow B'(-5, 4)$ STEP 3 — PLOT AND LABEL THE IMAGE. Plot $K'(-2, 1)$, $N'(-2, 4)$, and $B'(-5, 4)$ on the same grid. Join them to form triangle $K'N'B'$. Label the image clearly using prime notation (the dash symbol '). STEP 4 — CHECK YOUR WORK. – Measure the side lengths of the object and the image — they must be equal (isometry). – Check that every image point is the same distance from the origin as the corresponding object point: distance of K from origin $= \sqrt{1^2 + 2^2} = \sqrt{5}$; distance of K' from origin $= \sqrt{(-2)^2 + 1^2} = \sqrt{5}$. ✓ – The image should look like the object has been turned, not stretched or flipped. STEP 5 — STATE THE TRANSFORMATION FULLY. Always describe the transformation with all four pillars: 'Triangle KNB has been rotated 90° anticlockwise about the origin to give triangle K'N'B'.' Just as the minute hand of the Nairobi clock remains the same length no matter where it points, your triangle remains the same shape and size no matter how far it has been rotated.
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Section 4 — Finding the Centre, Angle, and Direction of an Unknown Rotation

Explain how you would work out the full description of a rotation when you are only given the object and the image (and the centre is not told to you). Describe at least two methods you can use. Explain why this skill matters.	Sometimes you are shown a shape and its rotated image, and you must figure out: Where is the centre? How many degrees? Which direction? This is like being handed two photographs of the Nairobi clock — one at 12:00 and one at an unknown time — and being asked to work out exactly how much time has passed. METHOD 1 — PERPENDICULAR BISECTORS (for finding the centre). The centre of rotation is always equidistant from each object point and its corresponding image point. This means it lies on the perpendicular bisector of the line segment joining each object vertex to its image vertex. – Draw the line segment from A to A' (object vertex to image vertex). – Construct the perpendicular bisector of AA'. – Repeat for the segment from B to B'. – The two perpendicular bisectors intersect at the centre of rotation. Once you have the centre, measure the angle from the centre to any object point, then to its image point — this gives you the angle of rotation.
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	METHOD 2 — TRACING PAPER (a practical method). – Trace the object shape onto tracing paper. – Place a pencil point on the suspected centre. – Rotate the tracing paper until the traced object matches the image. – The angle you turned through is the angle of rotation; the direction you turned is the direction of rotation. STATING THE DIRECTION. Once you know the angle, check whether the image is clockwise or anticlockwise from the object. A useful check: if the angle measured clockwise from object to image is θ, then anticlockwise it is $(360^\circ - \theta)$. Both describe the same rotation, so always pick the angle $\leq 180^\circ$ and state the correct direction, unless the question specifies otherwise. WHY THIS MATTERS IN REAL LIFE. Engineers designing the gear systems in Kenyan textile factories in Thika, or architects planning the rotating restaurant at the top of the Kenyatta International Convention Centre, must be able to reverse-engineer rotation — to look at a final position and calculate exactly what transformation produced it. The mathematics you are learning here is the foundation of that work.
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Section 5 — Answering the Driving Question: Connecting It All Back to the Nairobi Clock	
Now answer the driving question directly and completely. Your answer should bring together all four pillars of rotation, the coordinate rules, the method for rotating a shape, and the real-world relevance. Write at least two well-developed paragraphs. Imagine you are explaining to a fellow student who has never studied rotation before.	When the minute hand of the clock at the University of Nairobi completes its journey around the clock face, four mathematical rules govern exactly where every point on the hand ends up. First, every point rotates around one fixed centre — the pivot at the middle of the clock face, which on a coordinate grid we model as the origin (0, 0). Second, every point on the hand turns through the same angle — 90° for every 15 minutes, and a full 360° for one complete hour. Third, the direction matters: the hand moves clockwise, meaning from the top (12) toward the right (3), downward (6), and back to the top. Fourth, and most importantly, the hand is perfectly preserved — it never changes length, shape, or size, no matter where it points. These four pillars — centre, angle, direction, and preservation — are the complete mathematical description of rotation as a geometric transformation. To use these rules on a coordinate grid, you apply the rotation formulae to every point of the shape. For a 90° clockwise rotation about the origin, the rule is $(x, y) \rightarrow (y, -x)$. For 90° anticlockwise, it is $(x, y) \rightarrow (-y, x)$. For 180°, it is $(x, y) \rightarrow (-x, -y)$. You apply whichever rule matches your angle and direction to every vertex of the shape, plot the new points, join them, and label them with prime notation. The result is an image that is exactly congruent to the original — same side lengths, same angles — just turned to a new position. If the centre of rotation is not the origin, you translate the shape first, apply the rule, then translate back. To work backwards — finding the centre and angle when only the object and image are given — you construct perpendicular bisectors of the segments joining each vertex to its image, and measure the angle at their intersection. Every time you carry out any of these steps, you are doing exactly what the Nairobi clock does every second: turning a shape through a precise angle about a fixed point, landing it in a perfectly predictable new position, without changing a single measurement within it.

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Understanding of the Four Pillars of Rotation	All four pillars (centre, angle, direction, preservation/isometry) are correctly	All four pillars are named and mostly correctly explained. The Nairobi clock is	Fewer than three pillars are addressed, or at least one pillar contains a significant

	defined and explained with precise mathematical language. The Nairobi clock phenomenon is used naturally and accurately to illustrate every pillar. No misconceptions are present.	referenced for at least two pillars. Minor imprecision in language but no fundamental misconceptions.	misconception (e.g., stating that rotation changes the size of a shape, or omitting direction). The anchor phenomenon is not connected to the explanation.
Correct Application of Coordinate Rotation Rules	All three rotation rules (90° CW, 90° ACW, 180°) are stated correctly in (x, y) notation and applied accurately to at least one example. Working is shown clearly and all image coordinates are correct. The student explains why the rules preserve distance from the origin.	At least two of the three rules are stated and applied correctly. One computational error may be present but the method is sound. The student acknowledges that the rules apply when the centre is the origin.	Rules are stated incorrectly or applied with repeated errors. The student cannot distinguish between clockwise and anticlockwise rules, or applies a rule to the wrong coordinates. No checking of results is attempted.
Step-by-Step Method for Rotating a Shape on a Grid	A complete, logical sequence of steps is described: plotting the object, applying the rule to every vertex, plotting and labelling the image with prime notation, and checking results by comparing distances. All vertices are correctly transformed and the full transformation is described in a single complete statement naming all four pillars.	Most steps are present and correct. The student applies the rule to all vertices and labels the image. The final transformation statement may be incomplete (e.g., missing direction or centre). One plotting or labelling error may be present.	Steps are incomplete or out of order. The student applies the rule to fewer than all vertices, or plots image points incorrectly. No checking is done and the transformation is not described as a complete statement.
Finding an Unknown Rotation (Reverse Engineering)	At least two valid methods (e.g., perpendicular bisectors and tracing paper / angle measurement) are correctly described. The student explains how to determine both the centre and the angle, and correctly addresses how to determine direction. A real-world or Kenyan-context reason for this skill is given.	One valid method is clearly described and would lead to a correct answer. The student identifies the centre and angle but may be vague about determining direction. Some awareness of real-world relevance is shown.	The method described is incorrect or too vague to produce a result (e.g., 'guess and check' with no geometric reasoning). The student cannot clearly explain how to find the centre or measure the angle of an unknown rotation.
Quality of the Final Driving Question Response	The response answers the driving question directly, completely, and coherently. All four pillars, the coordinate rules, the rotation method, and real-world relevance are integrated fluently. The Nairobi clock phenomenon is used as a thread throughout. Language is precise, student-accessible, and free of major errors. Response is at least two well-developed paragraphs.	The response addresses the driving question and covers most key ideas. The anchor phenomenon is referenced. The writing is clear but some ideas are underdeveloped or loosely connected. At least one well-developed paragraph is present.	The response only partially addresses the driving question. Key ideas (e.g., coordinate rules or the four pillars) are missing or incorrect. The anchor phenomenon is absent or mentioned only superficially. The response reads as a list of facts rather than a coherent explanation.